# tinyAM :: CHEATSHEET

Formula-driven age-structured assessment

## 1 Setup tidy inputs

Each variable has a column; each row describes a year and age. Survey rows also identify the survey. Keep complete catch, biology and population-covariate grids; survey rows may be sparse.

catch

| year | age | obs | unit |
|---|---|---|---|
| 1983 | 2 | 18 | thousand fish |
| 1983 | 3 | 2585 | thousand fish |
| ⋮ | ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 0 | thousand fish |
| 2024 | 20 | 0 | thousand fish |

index

| year | age | survey | samp_time | obs | q_block | unit |
|---|---|---|---|---|---|---|
| 1983 | 2 | Fall 2J3KL | 0.8 | 698822 | 2 | thousand fish |
| 1983 | 3 | Fall 2J3KL | 0.8 | 517528 | 3 | thousand fish |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| 2024 | 19 | Fall 2J3KL | 0.8 | 0.0 | 6-20 | thousand fish |
| 2024 | 20 | Fall 2J3KL | 0.8 | 0.0 | 6-20 | thousand fish |

weight

| year | age | obs | unit |
|---|---|---|---|
| 1983 | 2 | 0.0704 | kg |
| 1983 | 3 | 0.201 | kg |
| ⋮ | ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 12.9 | kg |
| 2024 | 20 | 14.1 | kg |

maturity

| year | age | obs |
|---|---|---|
| 1983 | 2 | 0.00003 |
| 1983 | 3 | 0.00486 |
| ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 1.00000 |
| 2024 | 20 | 1.00000 |

pop_covariates

| year | age | NCAM_M_assumption |
|---|---|---|
| 1983 | 2 | 0.43 |
| 1983 | 3 | 0.38 |
| ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 0.36 |
| 2024 | 20 | 0.36 |

pop_covariates is optional.

```
dat <- make_tam_data(
  catch = catch_data, index = index_data,
  weight = weight_data, maturity = maturity_data,
  pop_covariates = population_data
)
```

The helper validates the tables. Plain lists also work: use **check_tam_data(dat)**; as_tam_data(dat) adds compact printing.

Label catch/index/weight units. Maturity is a proportion; sampling and spawning times are year fractions. Catch/index may contain NA; weight/maturity cannot. Population covariates need no obs column.

## 2 Specify model hypotheses

### Base: constant natural mortality

Fishing varies; natural mortality is constant. These are example settings, not package defaults.

```
base <- fit_tam(
  dat, years = 1983:2024, ages = 2:14,
  N_settings = list(process = "off", init = "exp"),
  rec_settings = list(process = "rw"),
  F_settings = list(process = "ar1", mu_form = ~ 1),
  M_settings = list(process = "off", mu_supplied = ~ I(0.3)),
  catch_settings = list(sd_form = ~ 1),
  index_settings = list(sd_form = ~ 1, q_form = ~ mono(q_block))
)
```

### Alternative survival assumptions

```
N_iid <- update(base,
  N_settings = list(process = "iid"))
M_ar1 <- update(base,
  M_settings = list(process = "ar1",
    age_breaks = c(2, 4, 6, 8, 14)))
```

**N IID:** apparent survival varies. Unexplained cohort changes may reflect mortality or migration; their causes are not separately estimated.

**M AR1:** fishing and natural mortality vary; M departures are shared within age blocks. Migration is considered negligible.

```
fits <- list(Base = base, N_iid = N_iid, M_ar1 = M_ar1)
```

Partial updates retain other settings. Use check_tam() to inspect numerical diagnostics and statistical support before interpreting results.

## 3 Visualize and compare

### Explore model fits

```
vis_tam(model_list = fits,
        output_file = "output/tinyAM_dashboard.
    html",
        open_file = FALSE)
```

**Online dashboard** · Local dashboard. Fishery → Numbers-at-age and Residuals.

### Test survey forecast skill

```
hind <- lapply(fits, fit_hindcast, folds = 20)
```

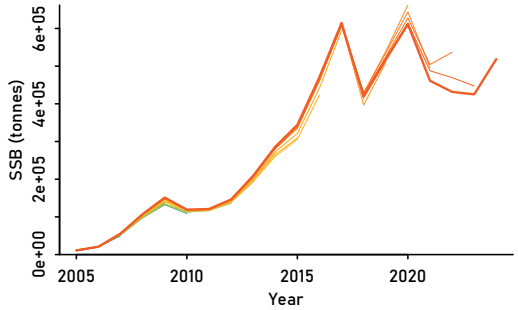
Returns retrospective summaries and one-year-ahead survey forecast scores, conditional on observed catch biomass.

### Survey forecast log-RMSE

| Model | RMSE | Folds | Rank |
|---|---|---|---|
| M AR1 | 0.765 | 18/20 | 1 |
| Base | 0.774 | 18/20 | 2 |
| N IID* | 0.784 | 2/20 | – |

Lower is better. Rank only complete, numerically acceptable runs with identical forecast observations. Folds count scored years. * Main fit did not converge.

### SSB stability: latest 20 years




M AR1 historical peels illustrate stability. RMSE scores survey observations given actual removals; it does not score SSB.

# tinyAM :: MODEL REFERENCE

Settings, equations and the population assumptions they represent

## Abundance and cohort survival

`N_settings`: `process = "iid"` allows cohort survival departures; `init = "exp"` initializes older ages. $Z = F + M$; recruitment $R_t = N_{t,1}$. Age indices $1, \ldots, A$ run from the youngest modeled age to the plus group; N is start-of-year abundance.

$$\begin{aligned}\widehat{N}_{t,a} &= N_{t-1,a-1}e^{-Z_{t-1,a-1}},\\ \widehat{N}_{t,A} &= N_{t-1,A-1}e^{-Z_{t-1,A-1}}\\ &\quad + N_{t-1,A}e^{-Z_{t-1,A}},\\ \log N_{t,a} &= \log \widehat{N}_{t,a} + \eta^N_{t,a}.\end{aligned}$$

For $t \geq 2$, non-plus ages use the first survival equation. Initial older ages use `"exp"` survivorship, `"free"` fixed states, `"latent"` integrated states, `"mono"` decline, or `"capped"` by first recruitment.

## Recruitment

`rec_settings`: `process = "rw"` gives a random walk. With IID/AR1 residuals, `mu_form = ~ bh(ssb)` or `~ ricker(ssb)` adds a stock–recruit curve.

$$\begin{aligned}\text{RW: } \log R_t - \log R_{t-1} &= \epsilon_t,\\ \log R_t &= \log g(SSB_{t-L}) + u_t,\\ g_{BH}(S) &= \alpha S/(1+\beta S),\\ g_{Ricker}(S) &= \alpha S e^{-\beta S}.\end{aligned}$$

$\alpha, \beta > 0$; $\epsilon_t \sim \mathcal{N}(0, \sigma_R^2)$. IID has $u_t = \epsilon_t$; AR1 has $u_t = \phi_R u_{t-1} + \epsilon_t$ and a stationary start. Curves describe medians. Lag $L$ defaults to the youngest biological age; early boundary R is free. No RW + curve.

## Biomass and spawning biomass

`maturity$spawn_time` supplies age/year-specific spawning fractions. Omit it for start-of-year SSB.

$$SSB_t = \sum_b n_{t,b} w_{t,b} p_{t,b} e^{-\tau_{t,b} Z_{t,b}}.$$

Biomass is $\sum_b n_{t,b} w_{t,b}$. $b$: retained biological age; $n$: its abundance; $w$: weight; $p$: mature fraction; $\tau$: year fraction (default 0). Hidden ages retain biology/timing and share plus-age mortality.

## Mean fishing mortality

`F_settings`: IID/AR1 with `mu_form = ~ logistic(age)` gives a rising long-term mean.

$$\begin{aligned}S(a) &= \{1 + e^{-k(a-a_{50})}\}^{-1}, \quad k > 0,\\ \log F_{t,a} &= \beta_F + \log S(a) + \eta^F_{t,a}.\end{aligned}$$

`mu_form = ~ mono(age)` instead adds cumulative nonnegative log-mean steps. Both curves require IID/AR1; annual F can depart from the rising mean.

## Natural mortality

`M_settings`: `mu_supplied = ~ M0` gives the baseline; `process = "iid"` adds departures.

$$\log M_{t,a} = \log M^0_{t,a} + \eta^M_{t,a}.$$

`M0` comes from `pop_covariates`. M states share age blocks and start in year 2 by default; other cells retain the mean. `"off"` fixes M to its mean; `mu_form` adds log-mean effects.

## Variation in N, F and M

`sd_form = ~ factor(age)` estimates age-specific process SDs. F `process = "cor_rw"` correlates increments across age. Ordinary age-based SD formulas give $\sigma_a = \exp(X_a \beta_\sigma)$. N, F and M have separate parameters; shared SDs do not share states.

$$\begin{aligned}\text{IID: } \eta_{t,a} &\sim \mathcal{N}(0, \sigma_a^2),\\ \text{RW: } \eta_{t,a} - \eta_{t-1,a} &\sim \mathcal{N}(0, \sigma_a^2).\end{aligned}$$

AR1 stationary covariance (state/block distances):

$$\text{Cov}(\eta_{t,a}, \eta_{s,b}) = \frac{\sigma_a \sigma_b \phi_t^{|t-s|} \phi_a^{|a-b|}}{(1-\phi_t^2)(1-\phi_a^2)}.$$

F `"cor_rw"`: independent annual increment vectors with

$$\Sigma_{ab} = \sigma_a \sigma_b \rho^{|a-b|}, \quad -1 < \rho < 1.$$

RW starts are unpenalized. N/M `"off"` gives $\eta = 0$.

## Survey catchability

`index_settings`: `q_form = ~ mono(q_block)` gives non-decreasing q under the default log link.

$$\eta_{q,k} = \beta_q + \sum_{j<k} d_j, \quad d_j \geq 0, \qquad q_k = e^{\eta_{q,k}}.$$

Zero steps give exact plateaus; `~ q_block` leaves age patterns unrestricted. With `q_link = "logit"`:

$$q_k = \{1 + e^{-\eta_{q,k}}\}^{-1}.$$

`~ logistic(age)` multiplies inverse-link q by $S(a)$. `iid()`, `rw()` and `ar1()` can add Gaussian variation. q covariates come from `index`.

## Catch and survey observations

Catch/index `sd_form = ~ 1` shares an observation SD; `~ survey` allows survey differences.

$$\begin{aligned}\widetilde{C}_{t,a} &= N_{t,a} \frac{F_{t,a}}{Z_{t,a}} (1 - e^{-Z_{t,a}}),\\ \tilde{I}_i &= q_i N_{t_i,a_i} e^{-s_i Z_{t_i,a_i}},\\ \log Y_i &\sim \mathcal{N}(\log \widetilde{Y}_i, \sigma_i^2),\\ \sigma_i &= \sigma_i^0 \exp(X_{sd,i} \beta_{sd}).\end{aligned}$$

$s_i$: survey sampling fraction. $\sigma_i^0 = 1$ unless a baseline SD is provided. Gaussian formula effects can also modify log SD. Predictions are medians; zero/NA observations are excluded, with complete predictions.

### Check fit, uncertainty and predictive skill

`check_tam()` checks numerical and statistical diagnostics; `fit_hindcast()` evaluates survey forecasts given observed removals and retrospective stability. The joint likelihood includes active process, formula-effect and observation densities. RTMB integrates random effects by Laplace approximation; intervals use local curvature. Standardized residuals $(\log Y_i - \log \widetilde{Y}_i)/\sigma_i$ depend on fitted, shrunken states. Convergence does not establish identifiability. Hindcasts return log RMSE and retrospective Mohn's $\rho$.